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Collisionless Damping of a Large Amplitude Whistler Wave

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Nonlinear damping of electromagnetic waves propagating along a uniform magnetic field has been calculated by computing nonlinear trajectories for trapped and resonant untrapped particles while using linear theory to describe the rest. This procedure parallels O'Neil's calculation for electrostatic modes, and the results are qualitatively similar. After an initial linear damping, amplitude oscillations set in and the amplitude quickly approaches a finite constant value. The frequency of the amplitude oscillations is temperature dependent. Phase mixing results from the spread in both parallel and perpendicular velocities, giving rise to a more rapid approach to the asymptotic amplitude than for electrostatic modes.

I. INTRODUCTION

It is well known[1] that small amplitude longitudinal waves in a warm isotropic plasma are damped, even in the absence of collisions (Landau damping). Pradhan[2] and Scarf[3] have shown that a small amplitude transverse wave propagating along a uniform magnetic field (B_0) in such a plasma also exhibits collisionless damping.

Scarf[3] has attributed this damping to the existence of a strong interaction between the wave and a small group of particles in resonance with it (cyclotron resonance). When a small amplitude circularly polarized electromagnetic wave with frequency ω and wavenumber k propagates along B_0, each particle "feels" the wave electric field rotating in the transverse plane with the Doppler-shifted frequency $(\omega - kV_{\|})$. If this frequency is approximately equal to the gyration frequency eB_0/m of particles around the background magnetic field lines, the particle continuously gains or loses energy in the manner of an oscillator driven at its natural frequency. For other particles the power absorbed changes sign rapidly and little net energy is exchanged. With an isotropic distribution, linear theory predicts a net gain in the energy of the resonant particles, and since energy must be conserved, the wave is damped.

For a large amplitude wave this damping mechanism (and the linear theory) breaks down after a finite time. One may see this by carrying out a coordinate transformation to the wave frame. In this frame the wave electric field is transformed away, and one is left with static helical magnetic field lines composed of the uniform background magnetic field and the circularly polarized wave magnetic field. The kinetic energy of the particles is conserved in this frame, so that resonant particles increasing their perpendicular velocity component lose parallel velocity. This eventually causes them to "get out of resonance" with the wave. The total amount of energy these particles can absorb from the wave is clearly limited.

The saturation of Landau damping for electrostatic modes was demonstrated by O'Neil.[4] We shall follow a procedure similar to O'Neil's to investigate the damping of large amplitude whistler modes.

First, in Sec. II, we shall establish the basic equations governing the system and use the helical symmetry of the fields in the wave frame to cast these equations in a simple form. We then review the linear theory and calculate the damping coefficient by considering the balance between resonant particle energy and wave and nonresonant particle energy. We shall show that linear theory breaks down for resonant particles after a time $\tau_R = (k\Omega_1 \bar{V}_\perp)^{-1/2}$, where $\Omega_1 = eB_1/m$ and B_1 is the wave magnetic field, but remains valid for a much longer time for nonresonant particles.

Then in Sec. III, we investigate the motion of charged particles in the wave frame, with the assumption (to be justified *a posteriori*) that the wave amplitude remains nearly constant. This analysis parallels in many respects the work of other investigators.[5–9] One finds two classes of resonant particles: those for which the phase difference between the rotating perpendicular velocity vector and the wave magnetic field vector is bounded (trapped particles); and those for which this phase difference grows without bounds (untrapped particles). With the restriction that the wave magnetic field is much smaller than the uniform background field, convenient approximations of resonant particle orbits valid for times longer than the linear breakdown time will be developed and used to construct the distribution function in the resonant region of phase space.

Using this distribution function we shall calculate the rate of energy absorption by the resonant particles and, again invoking conservation of energy, obtain a time dependent damping rate. We shall find that for short times the wave damps as described by linear theory. Later the amplitude increases, and after a few amplitude oscillations an undamped wave results; the whistler analog of the electrostatic BGK[7,10] modes.

II. REVIEW OF THE LINEAR THEORY OF CYCLOTRON DAMPING

A. Basic Equations

Consider an electron cyclotron wave of slowly varying amplitude propagating parallel to a uniform magnetic field $B_0\hat{e}_z$. The wave fields are

$$\mathbf{B}_1(Z,t)=B_1(t)[\hat{e}_x\sin(kZ-\omega t)+\hat{e}_y\cos(kZ-\omega t)], \tag{1a}$$

$$\mathbf{E}_1(Z,t)=(\omega/k)B_1(t)[\hat{e}_x\cos(kZ-\omega t) - \hat{e}_y\sin(kZ-\omega t)+O(\dot{B}_1/\omega B_1)]. \tag{1b}$$

The damping mechanism is most clearly exhibited if we calculate $B_1(t)$ by observing that energy gained by the plasma in the laboratory frame must be lost by the wave. The energy conservation equation is

$$\frac{d}{dt}\left(\frac{1}{2}\int_0^\lambda(\epsilon_0|\mathbf{E}_1|^2+\mu_0^{-1}|\mathbf{B}|^2)dZ\right)+P_{\text{res}}+P_{\text{nonres}}=0, \tag{2}$$

where $P=\frac{1}{2}nmd\langle V^2\rangle/dt$. P_{res} is the power per wavelength (λ) absorbed by resonant particles, and P_{nonres} is the power absorbed by nonresonant particles.

To calculate $\langle V^2\rangle$ one needs the electron distribution function, which is obtained by solving the Vlasov equation. When $\omega/k<c$, it is convenient to transform away the electric field by working in a coordinate system moving with the wave, so that the plasma experiences only the field

$$\mathbf{B}(z,t)=B_1(t)(\hat{e}_x\sin kz+\hat{e}_y\cos kz)+B_0\hat{e}_z.$$

The Vlasov equation may then be written in the form

$$\frac{\partial f}{\partial t}+v_{\|}\frac{\partial f}{\partial z}-\Omega_1v_\perp\cos(kz+\psi)\left(\frac{\partial f}{\partial v_{\|}}\right)_v - \left(\Omega_0+\frac{\Omega_1v_{\|}}{v_\perp}\sin(kz+\psi)\right)\frac{\partial f}{\partial\psi}=0, \tag{3}$$

where

$$v_\perp\left(\frac{\partial f}{\partial v_{\|}}\right)_v\equiv v_\perp\frac{\partial f}{\partial v_{\|}}-v_{\|}\frac{\partial f}{\partial v_\perp},\qquad \psi=\tan^{-1}\left(\frac{v_y}{v_x}\right),$$

$$\Omega_1=eB_1/m,\qquad \Omega_0=eB_0/m.$$

Note that the coordinates z and ψ occur only in the combination $kz+\psi$ in the coefficients of (3). If we replace z and ψ by the new coordinates $\phi_\pm=\pm kz-\psi$ and set $\partial f/\partial\phi_+\equiv0$, (3) becomes

$$\frac{\partial f}{\partial t}-\left(kv_{\|}+\Omega_0+\Omega_1\frac{v_{\|}}{v_\perp}\sin\phi_-\right)\frac{\partial f}{\partial\phi_-} -\Omega_1v_\perp\cos\phi_-\left(\frac{\partial f}{\partial v_{\|}}\right)_v=0. \tag{4}$$

$\partial f/\partial\phi_+\equiv0$ implies that the distribution function at any point z_1 differs from that at another point z_2 only by a rotation in velocity space of magnitude $k(z_2-z_1)$.

B. Derivation of the Linear Damping Rate

When $\Omega_1(t)$ is not too large, one can represent the distribution function as the sum of a spatially homogeneous part $f_0(v_{\|},v_\perp)$ and a perturbation $f_1(v_{\|},v_\perp,\phi,t)\sim O(\Omega_1)$. (We drop the subscript on ϕ_-.) The linearized equation

$$\frac{\partial f_1}{dt}-(kv_{\|}+\Omega_0)\frac{\partial f_1}{\partial\phi}=\Omega_1(t)v_\perp\cos\phi\left(\frac{\partial f_0}{\partial v_{\|}}\right)_v$$

has the solution

$$f_1=f_{10}(v_{\|},v_\perp)\sin[\phi+(kv_{\|}+\Omega_0)t] +\int_0^t\Omega_1(t')v_\perp\cos[\phi+(kv_{\|}+\Omega_0)(t-t')]\left(\frac{\partial f_0}{\partial v_{\|}}\right)_v dt'. \tag{5}$$

The first term on the right contains the initial perturbation and phase mixes away rapidly in integrals. This term will henceforth be ignored.

We shall use this solution to calculate the power absorbed by resonant and nonresonant particles, then obtain the linear damping rate from (2). Written in terms of wave frame variables, the power per wavelength absorbed by the plasma in the lab frame is

$$P_{\text{total}}=\frac{nm}{2}\int_{-\infty}^{\infty}dv_{\|}\int_0^\infty v_\perp\,dv_\perp \times\int_0^{2\pi}d\psi\left[v_\perp^2+\left(v_{\|}+\frac{\omega}{k}\right)^2\right]\int_0^\lambda\frac{\partial f}{\partial t}dz.$$

Solving for $\partial f/\partial t$ in the unlinearized Vlasov equation (4), setting $f=f_0+f_1$ on the right-hand side, then integrating over z yields

$$\int_0^\lambda\frac{\partial f}{\partial t}dz=\frac{\pi\Omega_1}{k}\left[v_\perp\left(\frac{\partial}{\partial v_{\|}}\right)_v-\frac{v_{\|}}{v_\perp}\right] \times\int_0^t\Omega_1(t')v_\perp\cos[(kv_{\|}+\Omega_0)(t-t')]\left(\frac{\partial f_0}{\partial v_{\|}}\right)_v dt'$$

and, after integrating by parts,

$$P_{\text{total}}=-\frac{\pi^2nm\Omega_1\omega}{k^2}\left(\int_{\text{res}}dv_{\|}+\int_{\text{nonres}}dv_{\|}\right) \times\int_0^\infty dv_\perp v_\perp^3\left(\frac{\partial f_0}{\partial v_{\|}}\right)_v\int_0^t\Omega_1(t')\cos[(kv_{\|}+\Omega_0)(t-t')]dt'. \tag{6}$$

If $(\Omega_0-\omega)\gg kV_T\gg\dot{\Omega}_1/\Omega_1$, the time integral in (6) can be approximated for thermal particles $[v_{\|}=-\omega/k+O(V_T)]$ by integrating by parts twice and neglecting terms of order $\ddot{\Omega}_1/(kv_{\|}+\Omega_0)^3$;

$$\int_0^t\Omega_1(t')\cos[(kv_{\|}+\Omega_0)(t-t')]dt' =\frac{\Omega_1(0)\sin(kv_{\|}+\Omega_0)t}{(kv_{\|}+\Omega_0)} +\frac{\dot{\Omega}_1(t)-\dot{\Omega}_1(0)\cos(kv_{\|}+\Omega_0)t}{(kv_{\|}+\Omega_0)^2}+\cdots. \tag{7}$$

The oscillating terms in (7) phase mix away when integrated over $v_{||}$, for large t. Ignoring these terms and expanding the remaining term, we have

$$\int_0^t \Omega_1(t')\cos[(kv_{||}+\Omega_0)(t-t')]dt'$$

$$\approx \frac{\dot{\Omega}_1(t)}{(\Omega_0-\omega)^2}\left\{1-2\left(\frac{kv_{||}+\omega}{\Omega_0-\omega}\right)+O\left[\left(\frac{kV_T}{\Omega_0-\omega}\right)^2\right]\right\}. \quad (7')$$

If the unperturbed distribution function is isotropic in the lab frame,

$$\left(\frac{\partial f_0}{\partial v_{||}}\right)_v=\frac{\omega}{kv_\perp}\frac{\partial f_0}{\partial v_\perp}. \quad (8)$$

Inserting this and (7′) into (6) and integrating by parts, one obtains for the nonresonant contribution to the power

$$P_{\text{nonres}}=\frac{2\pi nm\omega^2\Omega_1\dot{\Omega}_1}{k^3(\Omega_0-\omega)^2}\left\{1+O\left[\left(\frac{kV_T}{\Omega_0-\omega}\right)^2\right]\right\}. \quad (9)$$

When $(kv_{||}+\Omega_0)\approx0$ (resonance) the right-hand side of Eq. (7) is singular and the expansion fails. For slow damping, the resonant contribution to the power can be calculated by approximating $\Omega_1(t)$ by a constant. Thus,

$$\int_0^t \Omega_1(t')\cos[(kv_{||}+\Omega_0)(t-t')]dt'\approx\frac{\Omega_1\sin(kv_{||}+\Omega_0)t}{(kv_{||}+\Omega_0)}.$$

For large t, the function on the right has a high narrow peak centered at $v_{||}=-\Omega_0/k$ and of width $\Delta v_{||}\sim1/kt$, and oscillates rapidly elsewhere. Approximating this function by $\pi\Omega_1\delta(kv_{||}+\Omega_0)$ and using (6) and (8) yields, after an integration by parts,

$$P_{\text{res}}\approx(2\pi^2nm\Omega_1^2\omega^2/k^4)F_0[(\omega-\Omega_0)/k], \quad (10)$$

where

$$F_0(V_{||})=\int_0^{2\pi}d\psi\int_0^\infty V_\perp\, dV_\perp f_0(V_{||},V_\perp). \quad (10')$$

Setting

$$\Omega_1(t)=\exp\left(-\int_0^t\gamma(t)dt\right)$$

and using (1a, b) and (9) in the energy conservation equation (2) yields

$$\gamma(t)=\frac{P_{\text{res}}(t)}{(2\pi\epsilon_0m^2\Omega_1^2/ke^2)\{(\omega/k)^2+c^2+[\omega^2\omega_p^2/k^2(\Omega_0-\omega)^2]\}}, \quad (11)$$

where ω_p is the electron plasma frequency and, with (10), the linear damping rate is

$$\gamma_L=\frac{\pi\omega_p^2\omega^2F_0[(\omega-\Omega_0)/k]}{k^3\{(\omega/k)^2+c^2+[\omega^2\omega_p^2/k^2(\Omega_0-\omega)^2]\}}.$$

C. Breakdown of the Linear Theory

The solution of the linearized Vlasov equation (5) contains terms which grow with time and eventually make $(\partial f_1/\partial v_{||})_v\sim(\partial f_0/\partial v_{||})_v$, at which time the linearization fails. If Ω_1 is assumed constant, f_1 may be expanded in powers of $(kv_{||}+\Omega_0)$ to yield, for resonant particles,

$$f_1=f_{10}\sin[\phi+(kv_{||}+\Omega_0)t]+\Omega_1v_\perp\left(\frac{\partial f_0}{\partial v_{||}}\right)_v$$
$$\times\{t\cos\phi-\tfrac{1}{2}t^2(kv_{||}+\Omega_0)\sin\phi+O[t^3(kv_{||}+\Omega_0)^2]\}$$

so that

$$\left.\frac{(\partial f_1/\partial v_{||})_v}{(\partial f_0/\partial v_{||})_v}\right|_{(kv_{||}+\Omega_0)\approx0}\sim k\Omega_1v_\perp t^2$$

for large t. This ratio is of order unity in the resonance region when $t\sim\tau_R$, where $\tau_R=(k\Omega_1\bar{v}_\perp)^{-1/2}$. Breakdown occurs in the nonresonant region when $\tau\sim\tau_{NR}$, where $\tau_{NR}=\langle(\Omega_0+kv_{||})/k\Omega_1v_\perp\rangle\approx(\Omega_0-\omega)\tau_R^2$.

III. COLLISIONLESS DAMPING OF A LARGE AMPLITUDE WHISTLER WAVE

A. Outline of the Nonlinear Theory

The loss in amplitude sustained by the wave before the breakdown of the linear theory at $t\sim\tau_R$ is given by $\ln[\Omega_1(\tau_R)/\Omega_1(0)]=-\gamma_L\tau_R$. We define a "large amplitude wave" as one for which

$$\gamma_L\tau_R\ll1\equiv\Omega_1\gg\gamma_L^2/k\bar{v}_\perp. \quad (12)$$

For such amplitudes nonlinear effects appear before the wave has had time to damp appreciably.

If we require that $\Omega_1\ll(\Omega_0-\omega)^2/k\bar{v}_\perp$, which is not inconsistent with (12), it follows that $\tau_{NR}=(\Omega_0-\omega)\tau_R^2\gg\tau_R$ so that linear theory provides a valid description of nonresonant particles long after its breakdown in the resonant region. In particular, the result

$$\gamma(t)=\frac{P_{\text{res}}(t)}{(2\pi\epsilon_0m^2\Omega_1^2/ke^2)\{(\omega/k)^2+c^2+[\omega^2\omega_p^2/k^2(\Omega_0-\omega)^2]\}}$$

[Eq. (11)], remains valid for $t\gg\tau_R$. We shall obtain the nonlinear damping coefficient by reconstructing the distribution function f in the resonant region of phase space, using this to calculate $P_{\text{res}}(t)$, and then applying (11).

To construct the distribution function, observe that if the unperturbed distribution is isotropic in the lab frame at $t=0$, i.e., if $f_0=f_0(v^2+2\omega v_{||}/k+\omega^2/k^2)$, then for all later times

$$f(\mathbf{r},\mathbf{v},t)=f_0[v_0^2(\mathbf{r},\mathbf{v},t)+(2\omega/k)v_{||0}(\mathbf{r},\mathbf{v},t)+(\omega^2/k^2)], \quad (13)$$

where v_0, $v_{||0}$ are the initial values of v and $v_{||}$ for a particle at $\mathbf{r}$, $\mathbf{v}$ at time t. (Again we ignore the initial

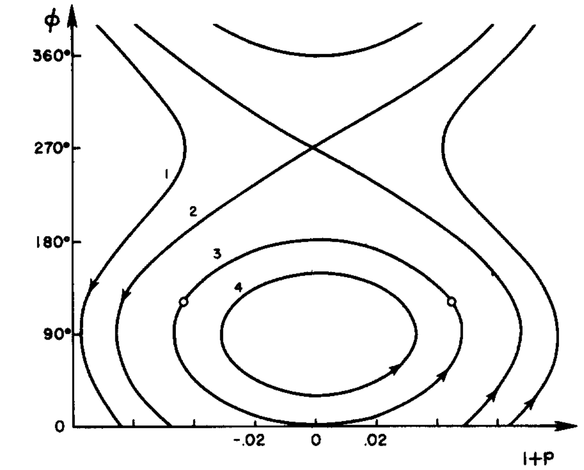

FIG. 1. Orbits in $(v_{||}, \phi)$ space for typical trapped and untrapped resonant particles $(p \equiv kv_{||}/\Omega_0)$. $B_1/B_0 = 0.001$, $kv/\Omega_0 = 1.5$, and $a(1) = 0.002$, $a(2) = 0.001118$, $a(3) = 0$, $a(4) = -0.0006$.

perturbation for large t.) In the next section we shall study the trajectories of individual particles and calculate v_0 and $v_{||0}$. In solving the equations of motion we shall treat the amplitude $\Omega_1 = eB_1/m$ as constant. This approximation will be justified *a posteriori*. (We know that B_1 changes only slightly before nonlinear effects appear, and we anticipate saturation of the damping mechanism in the nonlinear regime.)

B. Study of Particle Trajectories

The equation of motion in the wave frame is

$$m\frac{d\mathbf{v}}{dt} = -e\mathbf{v}\times\mathbf{B} \tag{14}$$

from which we obtain

$$\dot{v}_\perp = \Omega_1 v_{||}\cos\phi, \tag{14'}$$

$$\dot{v}_{||} = -\Omega_1 v_\perp \cos\phi, \tag{14''}$$

$$\dot{\phi} = -kv_{||} - \dot{\psi} = -[(kv_{||}+\Omega_0) + (\Omega_1 v_{||}/v_\perp)\sin\phi]. \tag{14'''}$$

It follows immediately from (14) that $\mathbf{v}\cdot\dot{\mathbf{v}} = v\dot{v} = 0$, so that the magnitude of the velocity vector is a constant of the motion.

$$v^2 = v_\perp^2 + v_{||}^2 = v_0^2. \tag{15}$$

A second constant of the motion has been discussed by several authors.[5–9] If we replace v, $v_{||}$, and Ω_1 by the dimensionless variables

$$p \equiv kv_{||}/\Omega_0, \qquad \Pi \equiv kv/\Omega_0, \qquad \epsilon \equiv \Omega_1/\Omega_0,$$

this constant may be written in the form

$$a = \tfrac{1}{2}(1+p)^2 - \epsilon(\Pi^2 - p^2)^{1/2}\sin\phi$$
$$= \tfrac{1}{2}(1+p_0)^2 - \epsilon(\Pi^2 - p_0^2)^{1/2}\sin\phi_0. \tag{16}$$

Equation (16) contains an important piece of information. The path of a particle in the two-dimensional phase space with coordinates p, ϕ can be determined by solving this equation for ϕ and plotting ϕ versus p. A typical set of such paths, with Π fixed, $\epsilon \ll 1$, and several values of a is illustrated in Fig. 1. For particles following the closed orbits the angle $90° - \phi$, which is the angle between $\mathbf{B}_1$ and $\mathbf{v}_\perp$, oscillates about zero. These particles are said to be "trapped" by the rotating wave fields.

Given the constants v and a it is possible to eliminate $v_\perp$ and ϕ from the $\dot{v}_{||}$ equation to form an integrable equation for $v_{||}$. After using (16) to obtain an expression for $\cos\phi$ and inserting this into (14''), we obtain

$$\left(\Omega_0^{-1}\frac{dp}{dt}\right)^2 + \Phi(p) = 0 \tag{17a}$$

or

$$\int \frac{dp}{[-\Phi(p)]^{1/2}} = \Omega_0 \int dt, \tag{17b}$$

where the "velocity potential" $\Phi(p)$ is

$$\Phi(p) = [\tfrac{1}{2}(1+p)^2 - a]^2 - \epsilon^2(\Pi^2 - 1) + \epsilon^2[(1+p)^2 - 2(1+p)]. \tag{18}$$

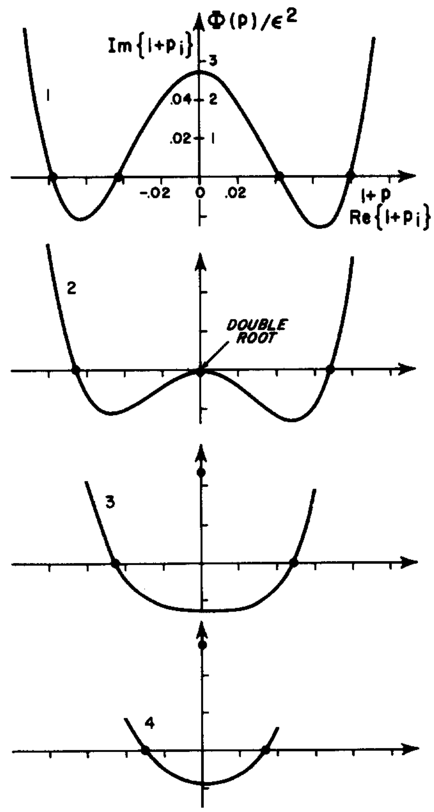

FIG. 2. Parallel velocity potential curves for particles in the orbits of Fig. 1. Motion is confined to the region where $\Phi(p) < 0$. The approximate roots of $\Phi(p_i) = 0$ are also shown. (In the case of conjugate complex pairs of roots, only the root with positive imaginary part is plotted.)

Figure 2 is a representation of the shape of the velocity potential $\Phi(p)$ for typical trapped and untrapped resonant particles (see also Refs. 6 and 7). On the same diagrams we have indicated the nature of the roots of $\Phi(p)$ and the region in which the motion is confined. The integrals in (17b) can be evaluated,[11] and exact expressions for $v_{||}(t)$ in terms of the roots of the polynomial $\Phi(p)$ have been presented by Roberts and Buchsbaum.[6] The exact roots of $\Phi(p)$ are quite complicated functions of the dynamical variables, however, and it is necessary to find workable approximate forms.

For this purpose we assume that $\epsilon \ll 1$. It follows from (16) that the first term on the right in (18) is of order ϵ^2. The second term is also of order ϵ^2, while the third term may be of higher order when $(1+p)$ is small, i.e., for resonant particles. If we set $(1+p_0)=0$ in (16), it becomes clear that $(1+p)\sim O(\epsilon^{1/2})$ for trapped particles. We shall consider untrapped particles to be resonant if they satisfy this same condition, i.e., if $(1+p)\sim O(\epsilon^{1/2})$. Thus, to lowest order in ϵ,

$$\Phi(p)\approx[\tfrac{1}{2}(1+p)^2-a]^2-\epsilon^2(\Pi^2-1) \tag{19}$$

and

$$a\approx\tfrac{1}{2}(1+p)^2-\epsilon(\Pi^2-1)^{1/2}\sin\phi. \tag{20}$$

With (19), (17b) may be integrated[11] and solved for p to yield

$$p\equiv(kv_{||}/\Omega_0)=-1+(2\sigma\eta/r)\,\mathrm{dn}[(\eta\Omega_0/r)(t_+-t),r] \tag{21a}$$

for untrapped resonant particles, and

$$p\equiv(kv_{||}/\Omega_0)=-1+(2\eta/r)\,\mathrm{cn}[\eta\Omega_0(t_+-t),1/r] \tag{21b}$$

for trapped particles, where

$$\begin{aligned}\sigma&=\mathrm{sgn}(1+p)\,,\ \eta(v)=\epsilon^{1/2}(\Pi^2-1)^{1/4},\\ r(a,v)&=2\eta(v)\,(2a+2\eta^2)^{-1/2}\\ &=2\eta(v)[(1+p)^2+2\eta^2(1-\sin\phi)]^{-1/2}\end{aligned} \tag{22}$$

and cn, dn are Jacobi elliptic functions. The parameter r is greater than one for trapped particles; has the value unity for particles on the separatrix; and is less than one for untrapped particles, with nonresonant particles having $r\sim\epsilon^{1/2}$. The phase constant t_+, which is the time at which the particle reaches its upper turning point, is conveniently expressed as a function of the dynamical variables by integrating (17b) as before and using (20) to eliminate p. After some algebraic manipulation we obtain

$$\begin{aligned}t_+&=t-(r/\Omega_0\eta)[\sigma F(\xi,r)+2\delta_{\sigma,-1}K(r)],\\ \xi&=\tfrac{1}{2}(\tfrac{1}{2}\pi-\phi)\end{aligned} \tag{23a}$$

for untrapped particles, and

$$\begin{aligned}t_+&=t-1/(\Omega_0\eta)[\sigma F(\theta,1/r)+2\delta_{\sigma,-1}K(1/r)],\\ \theta&=\sin^{-1}(r\sin\xi)\end{aligned} \tag{23b}$$

for trapped particles, where F is the elliptic integral of the first kind and $K(r)\equiv F(\tfrac{1}{2}\pi,r)$.

We are now able to express $v_{||0}$ as a function of the current dynamical variables. After setting $t=0$ in (21), substituting for t_+, and using the Fourier expansions of cn and dn, we obtain

$$\frac{kv_{||0}}{\Omega_0}+1=\frac{\pi\eta\sigma}{rK(r)}\left\{1+4\sum_{n=1}^{\infty}\frac{q_{(r)}{}^n}{1+q_{(r)}{}^{2n}}\times\cos\left[\frac{n\pi}{K(r)}\left(\frac{\eta\Omega_0t}{r}-\sigma F(\xi,r)\right)\right]\right\} \tag{24a}$$

for resonant untrapped particles, and

$$\frac{kv_{||0}}{\Omega_0}+1=\frac{4\pi\eta\sigma}{K(1/r)}\sum_{n=0}^{\infty}\frac{q_{(1/r)}{}^{n+1/2}}{1+q_{(1/r)}{}^{2n+1}}\times\cos\left(\frac{(2n+1)\pi}{2K(1/r)}[\eta\Omega_0t-\sigma F(\theta,1/r)]\right) \tag{24b}$$

for trapped particles, where $q(r)=\exp[-\pi K'(r)/K(r)]$ and $K'(r)=K[(1-r^2)^{1/2}]$.

Note that the coordinates $r, v, \xi(\phi)$ or $\theta(\phi)$ are degenerate in the sense that each point in r, v, ϕ space represents two points in the original phase space; e.g., the two small circles in Fig. 1 mark points with the same r, v, ϕ. This degeneracy is removed by the explicit change in the functional forms of $t_+(r, v, \xi)$, $v_{||0}(r, v, \xi)$, etc., which accompanies a change in the sign of $(1+p)$.

C. Calculation of the Resonant Power Absorption

With (13) and (15) the distribution function in the resonant region of phase space may be written in the form

$$f=f_0\left[v^2+\frac{\omega^2-2\omega\Omega_0}{k^2}+\frac{2\omega\Omega_0}{k^2}\left(\frac{kv_{||0}}{\Omega_0}+1\right)\right]$$

which we expand in the small quantity $kv_{||0}/\Omega_0+1$:

$$f=f_0\left(v^2+\frac{\omega^2-2\omega\Omega_0}{k^2}\right)+\frac{2\omega\Omega_0}{k^2}\left(\frac{kv_{||0}}{\Omega_0}+1\right)\frac{\partial f_0(V^2)}{\partial V^2}\bigg|_{V^2=[(\omega^2-2\omega\Omega_0)/k^2]+v^2}+\cdots. \tag{25}$$

The lab frame rate of energy absorption by resonant particles, expressed in terms of the wave frame variables, is obtained from f through the relation

$$P_{\mathrm{res}}(t)=\frac{d}{dt}\left[\frac{nm}{2}\iiint_{\mathrm{res}}d^3v\times\int_0^{\lambda}dz\left(v^2+\frac{2\omega}{k}v_{||}+\frac{\omega^2}{k^2}\right)f\right]. \tag{26}$$

The total wave frame kinetic energy of resonant

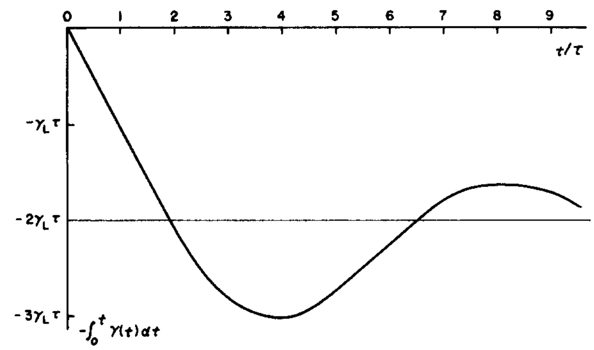

FIG. 3. The total logarithmic decrement in the wave amplitude vs time.

particles and the number of resonant particles per wavelength are conserved quantities. Thus, the integrals over $v^2 f$ and $(\omega^2/k^2)f$ are constants and vanish under the derivative in (26), which may be written in the form

$$P_{\rm res}(t)=\frac{nm\omega}{k}\frac{d}{dt}[I_T(t)+I_U(t)], \qquad (26')$$

where

$$I_T(t)=\iiint_{\text{trapped particles}} v_\perp\, dv_\perp\, dv_{||}\, d\psi \int dz\, v_{||} f,$$

$$I_U(t)=\iiint_{\text{untrapped particles}} v_\perp\, dv_\perp\, dv_{||}\, d\psi \int dz\, v_{||} f.$$

It is convenient to use r, v, ξ (or θ), z as integration variables in place of $v_\perp$, $v_{||}$, ψ, z:

$$I_T(t)=\frac{4\Omega_0}{k}\int_{\Omega_0/k}^{\infty} v\,dv \int_1^\infty \frac{dr}{r^3}\int_{-\pi/2}^{+\pi/2} d\theta\,\eta(v)\left(1-\frac{1}{r^2}\sin^2\theta\right)^{-1/2}$$

$$\times[v_{||<}(r,v,\theta)f_<(r,v,\theta,t)+v_{||>}f_>]\int_0^\lambda dz, \qquad (27a)$$

$$I_U(t)=\frac{4\Omega_0}{k}\int_{\Omega_0/k}^{\infty} v\,dv \int_0^1 \frac{dr}{r^3}\int_{-\pi/2}^{+\pi/2} d\xi\,\eta(v)\left(\frac{1}{r^2}-\sin^2\xi\right)^{-1/2}$$

$$\times(v_{||<}f_<+v_{||>}f_>)\int_0^\lambda dz, \qquad (27b)$$

where the subscripts $<$, $>$ imply $(1+p)<0$ and $(1+p)>0$, respectively. It is convenient to replace the variable r in (27a) by $1/r$. With this change, and after solving (22) for $v_{||}$, using (24) and (25), and discarding integrand terms odd in ξ, θ, we obtain

$$I_T(t)=\frac{2^8\pi^2\omega\Omega_0^3}{k^5}\sum_{n=0}^{\infty}\int_0^1 \frac{q_{(r)}^{n+1/2}\mathcal{G}_n^T(r)\Lambda(\alpha_n^T t)r\,dr}{K(r)(1+q_{(r)}^{2n+1})} + \text{const}, \qquad (28a)$$

$$I_U(t)=\frac{2^8\pi^2\omega\Omega_0^3}{k^5}\sum_{n=1}^{\infty}\int_0^1 \frac{q_{(r)}^{n}\mathcal{G}_n^U(r)\Lambda(\alpha_n^U t)\,dr}{r^4K(r)(1+q_{(r)}^{2n})} + \text{const}, \qquad (28b)$$

where

$$\mathcal{G}_n^T(r)=\int_{-\pi/2}^{+\pi/2}\cos\left((2n+1)\frac{\pi F(\theta,r)}{2K(r)}\right)\frac{r\cos\theta\,d\theta}{(1-r^2\sin^2\theta)^{1/2}}, \qquad (29a)$$

$$\mathcal{G}_n^U(r)=\int_{-\pi/2}^{+\pi/2}\cos\left(\frac{n\pi F(\xi,r)}{K(r)}\right)d\xi, \qquad (29b)$$

$$\Lambda(\alpha t)=\int_{\Omega_0/k}^{\infty}\eta^3(v)\left.\frac{\partial f_0(V^2)}{\partial V^2}\right|_{V^2=[(\omega^2-2\omega\Omega_0)/k^2]+v^2} \times\cos[\eta(v)\Omega_0\alpha t]dv, \qquad (29c)$$

$$\alpha_n^T=(2n+1)[\pi/2K(r)], \qquad \alpha_n^U=[n\pi/rK(r)].$$

The integrals over ξ, θ are evaluated in Appendix A. It is shown that

$$\mathcal{G}_n^T(r)=\frac{2\pi q_{(r)}^{n+1/2}}{1+q_{(r)}^{2n+1}}, \qquad \mathcal{G}_n^U(r)=\frac{2\pi q_{(r)}^{n}}{1+q_{(r)}^{2n}}. \qquad (30)$$

In Appendix B it is shown that when f_0 is Maxwellian, $\Lambda(\alpha t)$ is given approximately by the expression

$$\Lambda(\alpha t)\approx-\frac{2c_0\zeta_0}{c_1\pi^{1/2}}\frac{\Omega_1^2k^2}{\Omega_0^3}\tau F_0\left(\frac{\omega-\Omega_0}{k}\right)$$

$$\times\exp\left[-\left(\frac{\alpha}{2c_1\zeta_0}\frac{t}{\tau}\right)^2\right]\cos\left(\alpha\frac{t}{\tau}\right), \qquad (31)$$

where

$$\zeta_0\approx1.11, \qquad c_0\approx0.410, \qquad c_1\approx3.13,$$

and

$$\tau=(k\Omega_1V_T\zeta_0^2)^{-1/2}=[k(eB_1/m)V_T\zeta_0^2]^{-1/2}.$$

D. The Nonlinear Damping Coefficient

Combining (26′), (28), (30), and (31) we obtain for the power per wavelength absorbed by resonant particles

$$P_{\rm res}(t)=\left[\frac{2\pi^2nm\Omega_1^2\omega^2}{k^4}F_0\left(\frac{\omega-\Omega_0}{k}\right)\right]\frac{d}{d(t/\tau)}h\left(\frac{t}{\tau}\right), \qquad (32)$$

where

$$h\left(\frac{t}{\tau}\right)=2^9\pi^{1/2}\frac{c_0\zeta_0}{c_1}\int_0^1 dr\left(\sum_{n=1}^{\infty}\frac{1-\exp\{-[(n\pi/2c_1\zeta_0rK)(t/\tau)]^2\}\cos[(n\pi/rK)(t/\tau)]}{r^4K(r)(1+q^{2n})(1+q^{-2n})}\right.$$

$$\left.+\sum_{n=0}^{\infty}\frac{[1-\exp(-\{[(2n+1)\pi/4c_1\zeta_0K](t/\tau)\}^2)\cos\{[(2n+1)\pi/2K](t/\tau)\}]r}{K(r)(1+q^{2n+1})(1+q^{-2n-1})}\right). \qquad (33)$$

The quantity in square brackets in (32) is the result obtained for $P_{\rm res}$ from the linear theory. Thus, using (11), the damping rate is

$$\gamma(t)=\gamma_L h'\left(\frac{t}{\tau}\right)=\gamma_L\tau\frac{dh}{dt}, \tag{34}$$

where γ_L is the linear damping rate. The logarithmic decrement in the wave amplitude is

$$\ln\left(\frac{B_1(t)}{B_1(0)}\right)=-\int_0^t\gamma(t)dt=-\gamma_L\tau h\left(\frac{t}{\tau}\right). \tag{35}$$

In Fig. 3 the total logarithmic decrement is shown as a function of t/τ. Three features of this curve are noteworthy. For $t\lesssim 2\tau\approx 2\tau_R$ the slope of the curve is nearly constant and approximately equal to the value predicted by linear theory. For $t\gtrsim 2\tau$, Fig. 3 shows that the amplitude undergoes a damped oscillation with a frequency of approximately $1/(2\pi\tau)$. Since $\tau=(k\Omega_1V_T\zeta_0^2)^{-1/2}$, this frequency depends on the wavelength and amplitude of the wave, and on the temperature of the plasma. Finally, note that the amplitude oscillates about and decays toward a nonzero asymptotic value. From (33) and (35)

$$\ln\left(\frac{B_1(\infty)}{B_1(0)}\right)=-2^9\pi^{1/2}\frac{c_0\zeta_0}{c_1}\gamma_L\tau\int_0^1 dr$$
$$\times\left(\sum_{n=1}^{\infty}\frac{1}{r^4K(1+q^{2n})(1+q^{-2n})}+\sum_{n=0}^{\infty}\frac{r}{K(1+q^{2n+1})(1+q^{-2n-1})}\right).$$

The series appearing above have been summed by O'Neil.[4] We find that

$$\ln[B_1(\infty)/B_1(0)]\approx-2\gamma_L\tau.$$

In Fig. 4 the contributions of trapped and untrapped resonant particles to the logarithmic decrement are shown separately.

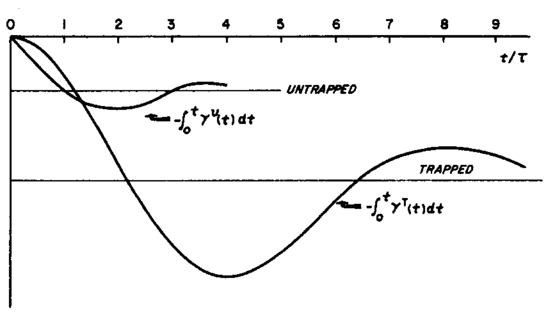

FIG. 4. The contributions of trapped and untrapped particles to the damping. $\gamma=\gamma^T+\gamma^U$.

E. Concluding Remarks

To interpret these results physically, recall that the kinetic energy of a particle, as measured in the lab frame, is

$$\text{K.E.}=\tfrac{1}{2}m[v^2+2(\omega/k)v_{\|}+(\omega^2/k^2)].$$

For resonant particles this energy oscillates with period T, which from (21) is

$$T\sim 4K(r)/\eta(v)\Omega_0=4K/(k\Omega_1\hat{v}_\perp)^{1/2},$$

where $\hat{v}_\perp=[v^2-(\Omega_0/k)^2]^{1/2}$. The average period for resonant particles is thus of order τ. For times $t\ll\tau$, particles gain or lose energy monotonically, with more particles gaining than losing if the unperturbed distribution is isotropic; exponential damping results. When t reaches a typical half-period, most of the particles which were initially absorbing energy are emitting, and the wave amplitude begins to oscillate. The spectrum of oscillation frequencies of individual particles is very broad, however, and after a few τ the phases are sufficiently mixed so that there is no net emission or absorption of energy by resonant particles.

The nonlinear cyclotron damping rate given by (33) and (34) is similar in form to that predicted for large amplitude electrostatic waves[4]; the most important difference being the appearance in $h(t/\tau)$, but not in its electrostatic counterpart, of the factors $\exp[-(\ )t^2]$. The similarity is explained by comparing (14b), (16), and (17) in the approximate form

$$\frac{d}{dt}(v_{\|}+\Omega_0)\approx-\frac{e}{m}(B_1\hat{v}_\perp)\cos\phi,$$

$$w=\left(\frac{\Omega_0}{k}\right)^2 a\approx\frac{1}{2}\left(v_{\|}+\frac{\Omega_0}{k}\right)^2-\frac{e(B_1\hat{v}_\perp)}{mk}\sin\phi,$$

$$\frac{d}{dt}(v_{\|}+\Omega_0)\approx\left[\left(\frac{e}{m}\right)^2(B_1\hat{v}_\perp)^2-k^2w^2+k^2w\left(v_{\|}+\frac{\Omega_0}{k}\right)^2-\tfrac{1}{4}k^2\left(v_{\|}+\frac{\Omega_0}{k}\right)^4\right]^{1/2}$$

with the corresponding equations for a particle moving in an electrostatic wave of "amplitude" $(B_1\hat{v}_\perp)$. The variation of $\hat{v}_\perp$ from particle to particle enhances the phase mixing and accounts for the factors $\exp[-(\ )t^2]$.

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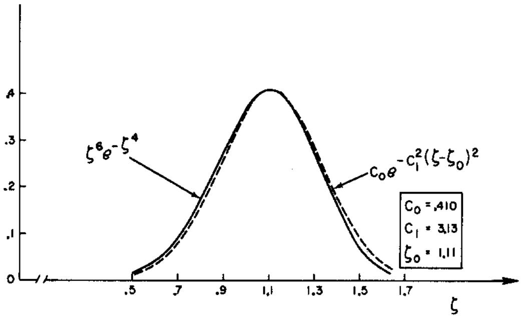

Fig. 5. Comparison of $\zeta^6 \exp(-\zeta^4)$ with its Gaussian approximation.

APPENDIX A. EVALUATION OF $\mathcal{g}_n{}^T(r)$, $\mathcal{g}_n{}^U(r)$

With the help of the identity $\cos\theta \equiv \mathrm{cn}[F(\theta, r), r]$, the transformation $u = F(\theta, r)$ puts (29a) in the form

$$\mathcal{g}_n{}^T(r) = r\int_{-K(r)}^{+K(r)} \mathrm{cn}(u,r)\cos\left((2n+1)\frac{\pi u}{2K(r)}\right)du$$

$$= \tfrac{1}{2}r\int_{-K}^{+K}\mathrm{cn}(u,r)\left[\exp\left(i\frac{(2n+1)\pi u}{2K}\right) + \exp\left(-i\frac{(2n+1)\pi u}{2K}\right)\right]du. \quad \text{(A1)}$$

It follows from the periodic properties of cn that

$$\int_{-K}^{+K}\mathrm{cn}(u,r)\exp\left(i\frac{N\pi u}{2K}\right)du = (1+q_{(r)}{}^N)^{-1}\int_C \mathrm{cn}(u,r)\exp\left(i\frac{N\pi u}{2K}\right)du,$$

where the contour C is a rectangle in the complex plane with corners at $\pm K(r), \pm K(r)+2iK'(r)$. $\mathrm{cn}(u, r)$ has a simple pole at $iK'(r)$ with residue $-i/r$. The residue theorem yields

$$\int_{-K}^{+K}\mathrm{cn}(u,r)\exp\left(i\frac{N\pi u}{2K}\right)du = \frac{2\pi}{r}\frac{q_{(r)}{}^{N/2}}{1+q_{(r)}{}^N}.$$

Applying this result to (A1), we obtain

$$\mathcal{g}_n{}^T(r) = 2\pi q^{n+1/2}/(1+q^{2n+1}). \quad \text{(A2)}$$

In similar fashion the transformation $u' = F(\xi, r)$ leads to the result

$$\mathcal{g}_n{}^U(r) = 2\pi q^n/(1+q^{2n}). \quad \text{(A3)}$$

APPENDIX B. APPROXIMATION OF $\Lambda(\alpha t)$

If we choose

$$f_0(V^2) = [V_T(\pi)^{1/2}]^{-3}\exp(-V^2/V_T{}^2),$$

and introduce the variable

$$\zeta = \left(\frac{v^2 - \Omega_0{}^2/k^2}{V_T{}^2}\right)^{1/4},$$

(29c) becomes

$$\Lambda(\alpha t) = -2\left(\frac{\Omega_1 k}{\pi\Omega_0{}^2 V_T}\right)^{3/2}\exp\left[-\left(\frac{\omega-\Omega_0}{kV_T}\right)^2\right] \times \int_0^\infty \zeta^6\exp(-\zeta^4)\cos[(k\Omega_1 V_T)^{1/2}\alpha t\zeta]d\zeta. \quad \text{(B1)}$$

The function $\zeta^6\exp(-\zeta^4)$ has a peak at $\zeta \approx 1.11$ and falls off rapidly on both sides of this point. Approximating this peak by a Gaussian and extending the integration to $-\infty$ yields

$$\int_0^\infty [\zeta^6\exp(-\zeta^4)]\cos(\chi\zeta)\,d\zeta \approx \int_{-\infty}^\infty \{c_0\exp[-c_1{}^2(\zeta-\zeta_0)^2]\}\cos(\chi\zeta)\,d\zeta = \frac{c_0}{c_1}\pi^{1/2}\exp[-(\chi/2c_1)^2]\cos(\chi\zeta_0), \quad \text{(B2)}$$

where $\chi = (k\Omega_1 V_T)^{1/2}\alpha t$ and $\zeta_0 = 1.11$. The Gaussian has the correct height and curvature at its peak if we choose $c_0 = 0.410$ and $c_1 = 3.13$.

In Fig. 5 the function $\zeta^6\exp(-\zeta^4)$ is graphically compared with its Gaussian approximation. With (10a), Eqs. (B1) and (B2) lead directly to the results quoted in (31).

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